

Scale Isolation and Chiral Orthogonality: Decoupling Acoustic Gradients from Transverse SU(2) Gauge Radiation in Geotopological Hydrodynamics

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ABSTRACT

We formalize the scale isolation and chiral orthogonality mechanism in Geotopological Hydrodynamics (GTH v12.0), proving how high-frequency longitudinal acoustic gradients are decoupled from transverse SU(2) gauge modes. We analyze the Chiral Orthogonality Governor $M_{G,eff}^2(x) = M_G^2 / (1 + Z(\nabla\phi))$, where $Z(\nabla\phi) \equiv I_{leak}^2 h^{ij} \nabla_i \phi \nabla_j \phi$, and demonstrate that in highly oscillatory or steep density gradient regimes ($I_{leak} |\nabla\phi| \gg 1$), the effective Stueckelberg gauge mass $M_{G,eff}^2 \rightarrow 0^+$. This suppresses unwanted longitudinal-to-transverse radiation leakage, preserving SU(2) gauge decoupling and guaranteeing compliance with solar system PPN bounds ($\gamma_{PPN} - 1.0 \leq 10^{-15}$) and binary pulsar orbital decay limits ($\delta P_b / P_b^{GR} \leq 10^{-14}$). We present the Clebsch fluid velocity decomposition $u_A = \nabla_A \phi + s \nabla_A \Theta + \alpha_l \nabla_A \beta_l + A_A^I q_I$, prove the positive definiteness of $M_{G,eff}^2$, and establish the exact pass/fail inequality $Z \geq \max(|C_1|/\epsilon_{\alpha 1}, |C_2|/\epsilon_{\alpha 2}, |C_P|/\epsilon_{P\Box}) - 1$ for preferred-frame leakage.

1. INTRODUCTION & THE SCALE ISOLATION CHALLENGE

In unified fluid-based spacetime theories, a persistent challenge is preventing high-frequency longitudinal acoustic modes (gravitational sector) from unphysically leaking into transverse gauge fields (electrodynamical/weak sectors). In Geotopological Hydrodynamics (GTH v12.0), this scale isolation is governed by the Chiral Orthogonality Governor, an intrinsic scale-separation mechanism embedded in the 5D substrate action.

The irreducible 7-coordinate constitutive tuple $\Theta = (M_{UV}, m_{IR}, \rho_0, K, G_{shear}, \tau_0, \eta_n)$ defines distinct propagation speeds for longitudinal compression ($c_s = \sqrt{K/\rho_0} = 1.2247 \times 10^8 \text{ m s}^{-1}$) and transverse shear ($c_{substrate} = \sqrt{G_{shear}/\rho_0} = 8.9443 \times 10^7 \text{ m s}^{-1}$). Decoupling these sectors requires that effective gauge field masses vanish in high-gradient regimes.

2. MATHEMATICAL FORMULATION OF THE CHIRAL GOVERNOR

The effective Stueckelberg mass of the projected SU(2) gauge connection is defined by:

$$M_{G,eff}^2(x) = \frac{M_G^2}{1 + Z(\nabla\phi)}, \quad Z(\nabla\phi) \equiv I_{leak}^2 h^{ij} \nabla_i \phi \nabla_j \phi \geq 0$$

THEOREM 1 (Governor Positivity and High-Gradient Suppression).

For any positive bare Stueckelberg mass $M_G^2 > 0$ and positive-definite spatial metric h^{ij} :

- (i) $M_{G,eff}^2(x) > 0$ strictly for all finite gradients $\nabla\phi$.
- (ii) As $I_{leak} |\nabla\phi| \rightarrow \infty$, $M_{G,eff}^2(x) \rightarrow 0^+$ asymptotically.

$$\lim_{I_{leak} |\nabla\phi| \rightarrow \infty} M_{G,eff}^2(x) = \mathcal{O}(|\nabla\phi|^{-2}) \longrightarrow 0^+$$

Proof: Since h^{ij} is positive definite, $Z(\nabla\phi) = I_{leak}^2 h^{ij} \nabla_i \phi \nabla_j \phi \geq 0$. Thus $1 + Z \geq 1 > 0$, which guarantees $M_{G,eff}^2 = M_G^2 / (1 + Z) > 0$. In highly oscillatory regions where $|\nabla\phi| \rightarrow \infty$, $Z \propto |\nabla\phi|^2 \rightarrow \infty$, forcing $M_{G,eff}^2 = \mathcal{O}(|\nabla\phi|^{-2}) \rightarrow 0^+$. Q.E.D.

3. CLEBSCH FLUID VELOCITY DECOMPOSITION

The 5D substrate velocity field u_A is decomposed using a multi-fluid Clebsch representation into potential background and non-Abelian circulation sectors:

$$u_A = \nabla_A \phi + s \nabla_A \Theta + \alpha_I \nabla_A \beta_I + A_A^I q_I, \quad I = 1, \dots, \dim \mathfrak{g}$$

On galactic and cosmological baselines, solenoidal flow is restricted to the force-free Beltrami branch $\nabla \times \mathbf{v}_B = \lambda_B \mathbf{v}_B$, ensuring that non-ideal stress remains transverse-orthogonal to normal dissipation:

$$\Pi_{\mu\nu} = f_n \Pi_{\mu\nu}^\eta + f_s \Pi_{\mu\nu}^s, \quad f_n(x) = \exp\left(-\frac{g_a(x)^2}{a_0^2}\right), \quad f_s(x) = 1 - f_n(x)$$

4. PREFERRED-FRAME LEAKAGE & BINARY PULSAR CONSTRAINTS

Longitudinal-to-transverse leakage parameter $\varepsilon_{\text{leak}}(x) \equiv 1 / (1 + Z(\nabla\phi))$ enters post-Newtonian parameters and binary timing decay rates:

$$\alpha_1 = C_1 \varepsilon_{\text{leak}} + \mathcal{O}(\varepsilon_{\text{leak}}^2), \quad \alpha_2 = C_2 \varepsilon_{\text{leak}} + \mathcal{O}(\varepsilon_{\text{leak}}^2), \quad \delta \dot{P}_b = C_P \dot{\varepsilon}_{\text{leak}}$$

THEOREM 2 (Preferred-Frame Admissibility Inequality).
For observational tolerances $\varepsilon_{\alpha 1}, \varepsilon_{\alpha 2}$ and $\varepsilon_{P\blacksquare}$ compliance requires:

$$Z \geq \max\left(\frac{|C_1|}{\varepsilon_{\alpha_1}}, \frac{|C_2|}{\varepsilon_{\alpha_2}}, \frac{|C_P|}{\varepsilon_{\dot{P}}} \right) - 1$$

5. GAUGE CURRENT CONSERVATION & SPIN-VORTICITY CODIFFERENTIAL

Contraction of the closed bulk 3-form $M^{(3)}$ with the normal n^A defines the boundary spin-vorticity bivector $S^{\nu\mu}$. The gauge current J_Q^μ is:

$$J_Q^\mu \equiv \nabla_\nu S^{\nu\mu}, \quad \nabla_\mu J_Q^\mu = \nabla_\mu \nabla_\nu S^{\nu\mu} = 0$$

Antisymmetry $S^{\nu\mu} = -S^{\mu\nu}$ guarantees exact, automatic gauge current conservation $\nabla_\mu J_Q^\mu = 0$.

6. CONCLUSION & MONOGRAPH SERIES ROADMAP

The Chiral Orthogonality Governor $M_{G,\text{eff}}^2 = M_G^2 / (1 + Z)$ decouples high-frequency longitudinal acoustic modes from SU(2) gauge fields, ensuring solar system PPN compliance ($\gamma_{\text{PPN}} - 1.0 \leq 10^{-15}$).

Paper #	Monograph Title / Physical Focus	Key Formulation
Paper 1	Density Ceiling & Core Regularization	$V_{\text{top}}(\rho) \rightarrow +\infty, R_c \geq (3M_c / 4\pi \rho_{\text{max}})^{1/3}$
Paper 2	Scale Isolation & Chiral Orthogonality	$M_{\{G,\text{eff}\}}^2 = M_G^2 / (1 + Z(\text{grad } \phi))$
Paper 3	Beltrami Vortex Wakes & Disk Dynamics	$\beta(r) = \chi_n G_{\text{model}} \Sigma_b(r) e^{-(\text{Re } GTH)}$
Paper 4	Acoustic Horizons & Echo Comb	$\tau_{\text{echo}}(\omega) = 2 (d \Phi_{\text{sv}} / d \omega)$
Paper 5	Zero-Mode Gauss-Codazzi Projection	$G^{(4)}_{\{\mu\nu\}} + \Lambda_{\text{curv}} h_{\{\mu\nu\}} = \kappa_{\text{SI}} T^{\wedge\{\text{eff}\}}_{\{\mu\nu\}} + C_{-\{\mu\nu\}}$